Machine Learning, Deep Learning & NLP Applications

Capstone Project Report

Fake News DetectionusingMachine Learning & Neural Networks

Zeyad Talal AL-Mashouq
Class103

1. Introduction

For this project, I chosethe Fake News Detection dataset because I was interested in learning how machine learning can beused to identify whether news articles are real or fake. This project provided an opportunity to applydata preprocessing, machine learning, and deep learning techniques to a real-world text classification problem.

2. Dataset Description

The dataset consists of news articles labeled as either real or fake. Since the dataset contains text data, several preprocessing steps were applied before training the models.

The preprocessing process included:

- Converting all text to lowercase.
  Removing punctuation and special characters.
- Removing common English stop words.
- Converting text into numerical features using TF-IDF vectorization.
-
-

After preprocessing, the dataset was divided into training and testing sets using an 80% training split and a 20% testing split.

3. Models Used

Three machine learning models were trained and evaluated:

- Logistic Regression
- Random Forest
- K-Nearest Neighbors (KNN)

The performance of each model was evaluated using Accuracy, Precision, Recall, and F1-score. Confusion matrices were also generated to better understand how each model performed on the classification task.

After comparing the results, one model achieved the strongest overall performance and was selected as the best traditional machine learning model for this project.

4. Neural Network

A neural network was developed using Keras to perform the same classification task. The network consisted of multiple Dense layers with ReLU activation functions and included a Dropout layer to help reduce overfitting.

The model was trained for multiple epochs, and both training and validation accuracy and loss were monitored throughout the training process. These learning curves helped evaluate whether the model was learning effectively and whether overfitting was occurring.

The neural network successfully learned patterns within the text data and produced competitive classification results.

5. Results and Comparison

The machine learning models and the neural network were evaluated using the same metrics to ensure a fair comparison.

The results showed that all models were able to classify news articles with a reasonable level of accuracy. However, some models performed better than others due to their ability to capture patterns in textual data.

The neural network demonstrated strong performance and provided an effective alternative to traditional machine learning approaches. While it required more training time and computational resources, it was able to learn more complex relationships within the data.

Overall, the comparison highlighted the advantages and limitations of both traditional machine learning algorithms and deep learning models for text classification tasks.

6. What I Learned

This project helped me gainpractical experience in building a complete machine learning pipeline from start to finish.

Some of the most important lessons I learned include:

- Data preprocessing plays a critical role in improving model performance.
- TF-IDF is an effective method for converting text into numerical features.
- Different machine learning algorithms can produce significantly different results on the same dataset.
- Neural networks are powerful tools for text classification but often require more training time.
- Model evaluation metrics such as Accuracy, Precision, Recall, and F1-score are essential for understanding performance.
- Building and comparing multiple models provides valuable insight into selecting the most appropriate solution for a problem.

In conclusion, this project successfully demonstrated the application of machine learning and deep learning techniques to fake news detection and provided valuable hands-on experience with real-world data analysis and classification.